

[Technical Summary] Mulberry Project: Voice-to-Table AI OS

Implementing Google AP2 for Inclusive AI Finance & Food Desert Relief

1. Project Objective

The **Mulberry Project** aims to solve "Food Deserts" in South Korea by providing a voice-based commerce ecosystem for the elderly. We utilize **Edge AI (DeepSeek)** and **Google AP2** to bridge natural language intent with secure financial execution.

2. Technical Architecture: The "Agentic Bridge"

The Mulberry OS acts as a mediator between the user's voice and the **NH (NongHyup) Bank Open API**.

- **Edge Layer:** On-device **DeepSeek-V3** processes dialect-heavy Korean speech, performing STT and Intent Extraction locally to ensure low latency (<800ms) and data privacy.
- **Protocol Layer:** Implementation of the **Google AP2 3-Stage Mandate System** to enforce financial boundaries for non-digital natives.
- **Settlement Layer:** Mapping AP2 payloads to **NH Bank's RESTful APIs** for real-time account-to-account (A2A) transfer or card billing via secure tokens.

3. AP2 Mandate Flow Implementation

We have strictly mapped our business logic to the official AP2 lifecycle:

1. Intent Mandate (User → Agent):

- **Constraints:** We apply `Max_Amount` (e.g., 50,000 KRW) and `MCC_Category_Lock` (5411 - Groceries).
- **Auth Proof:** We encapsulate **FIDO-based Voice Biometric Hashes** within the `Auth_Proof` field for non-repudiation.

2. Cart Mandate (Agent → Merchant):

- **Inventory Sync:** Real-time SKU and price validation with the **NH Hanaro Mart** inventory database.
- **Validation:** Prevents AI hallucinations regarding item availability or pricing.

3. Payment Mandate (Agent → Bank):

- **Tokenization:** AP2 `Payment_Mandate` triggers an **NH Open API** call using a pre-authorized **Billing Key**.

- **Security:** No sensitive financial data (PAN, CVV) is processed or stored within the voice interaction layer.

4. Key Engineering Challenges & Goals

- **Dialect Resilience:** Fine-tuning the Intent Extraction prompt to handle regional Korean dialects (e.g., Gangwon, Gyeongsang).
- **Low-Resource Optimization:** Running the mandate generation engine on low-power ARM-based Edge terminals.
- **Compliance:** Aligning with South Korea's strict financial data residency and "Zero-Trust" security frameworks.

5. Collaboration Request

We seek technical alignment with the **Google Agentic Commerce** community to:

- Validate our **Mandate-to-Local API mapping** strategy.
- Discuss **SDK optimizations** for Edge AI environments.
- Contribute to the **Global ESG Use-Case** library for AP2.

Lead Developer: re.eul (CEO, Mulberry Project)

Repository: [GitHub - wooriapt79/mulberry-](https://github.com/wooriapt79/mulberry-) [food desert project](#) **Declaration:** "Technology serving the most vulnerable."
